TV Intelligence Scanner — Hardware Report

Project #1 SAMPLE — Bench TV (synthetic data) | Acme TESTTV55 | generated 2026-11-08 22:07 UTC | v1.0.0

Every specification value below lists its source: 'datasheet' (text of a datasheet in the local library, with page), 'decoder' (vendor part-numbering scheme; confirm with datasheet), 'marking_rule' (numbering convention: vendor/family only) or 'user' (entered by the operator with a reference). Values without evidence are reported as not identified.

1. Motherboard summary

Mainboard	TP.MS338.PB801
PCB revision	not identified
Board part numbers	—
SoC / CPU	MStar (now MediaTek) MSD6A648
GPU	not identified (no evidence)
RAM	RAM: 1 GB total (from decoder + datasheet; confidence 90%) K4B4G1646E-BYK0 ×2: 512 MB each (decoder + datasheet)
Flash	flash: 8 MB total (from decoder; confidence 80%; confirm with datasheet) W25Q64JVSIQ ×1: 8 MB each (decoder)
Wi-Fi	not identified (no evidence)
Bluetooth	not identified (no evidence)
Wi-Fi / BT module	not identified (no evidence)
Audio	not identified (no evidence)
T-Con	not identified (no evidence)
Power supply	not identified (no evidence)
Display panel	LC550EGE

2. Components

#	Board/pos.	Category	Part number	Manufacturer	Qty	Identification	Conf.
3	mainboard U7	Flash memory	`W25Q64JVSIQ`	Winbond	1	Decoded from vendor part-numbering scheme	75%
2	mainboard U3	RAM	`K4B4G1646E-BYK0`	Samsung	2	Datasheet match (exact part number)	90%
1	other	SoC / CPU	`MSD6A648`	MStar (now MediaTek)	1	Vendor/family from numbering convention only	66%

3. Component details

#3 W25Q64JVSIQ — Flash memory

Attribute	Value	Source	Reference	Conf.
density_bits	64 Mbit	decoder	decoder winbond_spi_nor: Winbond SpiFlash naming (W25Q + density in Mbit)	80%
device_family	SPI flash	marking_rule	marking rule winbond_spi	75%
manufacturer	Winbond	decoder	decoder winbond_spi_nor: Winbond SpiFlash naming (W25Q + density in Mbit)	80%
manufacturer	Winbond	marking_rule	marking rule winbond_spi	75%
memory_type	SPI NOR flash	decoder	decoder winbond_spi_nor: Winbond SpiFlash naming (W25Q + density in Mbit)	80%

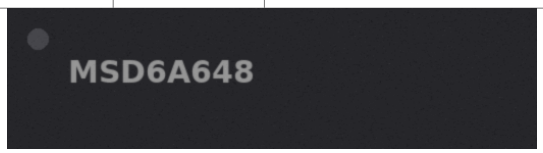
#2 K4B4G1646E-BYK0 — RAM

Attribute	Value	Source	Reference	Conf.
data_rate	1600 MT/s	datasheet	datasheet #1 p.1	60%
density_bits	4 Gbit	decoder	decoder samsung_ddr: Samsung DRAM part numbering (K4 + family + density + organisation) (corroborated by datasheet)	90%
density_bits	4 Gbit	datasheet	datasheet #1 p.1	60%
device_family	DRAM	marking_rule	marking rule samsung_dram	70%
manufacturer	Samsung	decoder	decoder samsung_ddr: Samsung DRAM part numbering (K4 + family + density + organisation)	80%
manufacturer	Samsung	marking_rule	marking rule samsung_dram	70%
memory_type	DDR3 SDRAM	decoder	decoder samsung_ddr: Samsung DRAM part numbering (K4 + family + density + organisation) (corroborated by datasheet)	90%
memory_type	DDR3	datasheet	datasheet #1 p.1	65%
organization	x16	decoder	decoder samsung_ddr: Samsung DRAM part numbering (K4 + family + density + organisation)	80%
package	96-BALL FBGA	datasheet	datasheet #1 p.1	65%
package	FBGA	datasheet	datasheet #1 p.2	65%
supply_voltage	1.5 V	datasheet	datasheet #1 p.1	65%

Datasheet #1: 4Gb DDR3 SDRAM test sheet (SHA-256 e1fdf9f79a7ffa66…; source: not recorded)

#1 MSD6A648 — SoC / CPU

Attribute	Value	Source	Reference	Conf.
device_family	TV SoC / scaler	marking_rule	marking rule mstar_soc	70%
manufacturer	MStar (now MediaTek)	marking_rule	marking rule mstar_soc	70%



Evidence #2 (chip.png, SHA-256 3b3be0c0b194b984…)